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Clinical efficacy of high versus low intensity diode laser as an adjunct to non-surgical periodontal therapy in periodontitis: randomized controlled clinical trial

Amani A. Al-Sharani[1,4]✉, Wadhah A. Al-Hajj[1,2] & Ahmed A. Madfa[3]

Scaling and root planing (SRP) is the standard nonsurgical treatment for periodontitis; however, its efficacy in eliminating subgingival biofilm and promoting periodontal healing is often limited, particularly in deep pockets. Adjunctive diode laser therapy has been proposed to enhance clinical outcomes through antimicrobial and host-modulatory effects. This study aimed to evaluate the clinical efficacy of high- and low-intensity diode laser therapy as adjuncts to SRP. A randomized, within-patient, site-level controlled clinical trial was conducted on 27 patients, encompassing 81 periodontal quadrants and 596 sites. Periodontal sites were randomly assigned to three treatment groups: SRP alone as the control, SRP combined with high-intensity diode laser therapy (DLT-H), and SRP combined with low-intensity diode laser therapy (DLT-L). Clinical indices—plaque index (PI), gingival index (GI), probing pocket depth (PPD), and clinical attachment level (CAL)—were assessed at baseline (T0), after one month (T1), and after three months (T2). The results were analyzed using repeated-measures ANOVA followed by post-hoc pairwise comparisons. All three treatment approaches led to statistically significant reductions in PI, GI, PPD, and CAL across the study period ($p < 0.001$). DLT-H showed greater short-term CAL gains, with significant improvement observed at T1 ($p = 0.001$ vs. SRP; $p = 0.037$ vs. DLT-L) and T2 ($p < 0.001$ vs. SRP; $p = 0.018$ vs. DLT-L). The DLT-L produced moderate clinical gains, whereas SRP alone yielded early improvements that plateaued over the follow-up period. In contrast, the DLT-H demonstrated consistently superior outcomes compared with both DLT-L and SRP, most notably in clinical attachment level (CAL), indicating more favorable clinical outcomes. Integrating high-intensity diode laser therapy as an adjunct to nonsurgical periodontitis management provides significant short-term clinical benefits, particularly in CAL improvement and inflammation control. In comparison, low-intensity diode laser application offers only moderate advantages, while traditional SRP alone remains effective yet demonstrates minimal periodontal healing capacity. These findings endorse the judicious application of diode laser therapy—especially high-intensity settings—to enhance periodontal outcomes and attachment gain. Long-term studies and further biological assessments and additional biological assessments are warranted to confirm these findings.

***Trial registration*: NCT07013162.**

Keywords Periodontitis, Scaling and root planing, Diode laser, Clinical attachment level, Nonsurgical therapy, Photobiomodulation

[1]Oral Medicine and Periodontology Department, College of Dentistry, University of Sciences and Technology, Aden, Yemen. [2]Department of Periodontology, Faculty of Dentistry, Department of Dentistry, Faculty of Medical Sciences, Thamar University, Civilization University, Dhamar, Yemen. [3]Department of Restorative Dental Science, College of Dentistry, University of Ha'il, Ha'il, Kingdom of Saudi Arabia. [4]Department of Biological and Preventive Sciences, College of Dentistry, University of Sciences and Technology, Sana'a, Yemen. ✉email: aamanialsharani.aa@gmail.com

Background

Periodontitis is an inflammatory condition with multiple contributing factors, characterized by the gradual destruction of the supporting structures of the teeth, including the periodontal ligament and alveolar bone. The disease is primarily driven by the buildup of bacterial biofilm, which provokes an excessive or dysregulated immune response in vulnerable individuals[1,2]. Clinically, it is marked by deeper periodontal pockets, gingival inflammation, loss of clinical attachment, and resorption of alveolar bone, potentially leading to tooth mobility and eventual tooth loss if not properly managed.

SRP continues to be the cornerstone of nonsurgical periodontal therapy[3], effectively removing subgingival biofilm and calculus through mechanical debridement. SRP consistently improves clinical outcomes, including PI, gingival index GI, probing pocket depth PPD, and CAL, particularly in moderately deep periodontal pockets[4,5]. However, mechanical treatment alone has notable limitations, especially in anatomically challenging areas such as deep pockets, furcations, and root concavities. Moreover, some periodontal pathogens can penetrate the soft tissues, rendering them less accessible to conventional debridement techniques.

To address the limitations of mechanical debridement, various adjunctive approaches have been investigated, including local antimicrobial agents, photodynamic therapy, and laser-assisted interventions. Diode lasers, in particular, have received considerable attention due to their ability to both reduce microbial burden and support tissue healing when used alongside SRP. Functioning within the 810–980 nm wavelength range, diode lasers can be applied in two main modes: high-intensity (DLT-H) and low-level/photobiomodulation (DLT-L). DLT-H produces photothermal effects that remove inflamed sulcular epithelium, reduce bacterial populations, and improve pocket decontamination, whereas DLT-L promotes cellular changes, angiogenesis, and collagen formation without generating harmful thermal effects[6,7].

Clinical evidence suggests that adding diode laser therapy to SRP can enhance treatment outcomes, resulting in more pronounced reductions in PPD and GI, as well as greater gains in CAL, compared with SRP alone[8,9]. DLT-H offers immediate benefits in deep pockets through superior microbial control and targeted tissue ablation, while DLT-L supports periodontal healing responses and modulation of gingival inflammation.

Recent clinical evidence demonstrates that diode lasers (λ 808–980 nm), when used adjunctively with SRP, can provide additional short-term improvements in gingival inflammation, PPD, and CAL[10–12]. Photobiomodulation mediated by low-level lasers has also been reported to accelerate tissue repair and reduce postoperative symptoms. High-power lasers, including Er: YAG and Er, Cr: YSGG, have shown efficacy in both surgical and non-surgical contexts, with significant improvements in PPD, CAL, GI, and bleeding on probing (BOP)[11,13,14]. Split-mouth randomized controlled trials (RCTs) indicate that adjunctive diode and erbium laser therapy can modulate immunological parameters, reducing matrix metalloproteinase-13 (MMP-13) levels and increasing interleukin-10 (IL-10) in gingival crevicular fluid[15].

Comparative studies have further highlighted the relative efficacy of different adjunctive approaches. For instance, Er, Cr: YSGG laser therapy produced superior improvements in PD, CAL, PI, and GI compared with a-PDT or SRP alone during supportive periodontal therapy[16]. Similarly, adjunctive antimicrobial photodynamic therapy using diode lasers has demonstrated clinical benefits, although results remain inconsistent across studies[17,18]. Variability in study outcomes is attributed to heterogeneity in laser wavelengths, power settings, irradiation protocols, treatment frequency, and methodological quality.

Despite these encouraging findings, the supporting evidence remains diverse. Reported protocols differ widely in wavelength, power settings, exposure duration, and treatment frequency, hindering the development of standardized clinical guidelines. Moreover, most investigations have assessed either DLT-H or DLT-L in isolation, and few well-designed trials have directly compared the two approaches under uniform conditions[19]. This lack of head-to-head data leaves clinicians uncertain about their relative advantages and the specific clinical indications for each modality.

To address these gaps, a direct evaluation of both DLT-H and DLT-L as adjuncts to conventional SRP is required. Therefore, this study assessed the clinical efficacy of both modalities under standardized conditions by comparing their effects on key periodontal parameters—PI, GI, PPD, and CAL—in patients with periodontitis. The null hypothesis posited that there would be no significant differences in PI, GI, PPD, or CAL among patients receiving SRP alone, SRP with DLT-H, or SRP with DLT-L.

Methods

Trial design and ethical considerations

This prospective, randomized, within-patient, split-site, controlled clinical trial was carried out over a three-month period at the Dental Polyclinics of the Faculty of Dentistry, University of Science and Technology (UST), Sana'a, Yemen. The study protocol received approval from the University Research Ethics Committee (UREC) of UST (Study ID: 1446/0016/UREC/UST) and adhered to the ethical principles of the Declaration of Helsinki as well as the CONSORT 2010 guidelines. The trial was registered at ClinicalTrials.gov (NCT07013162; first posted 24/05/2025). Written informed consent was obtained from all participants before enrollment, ensuring compliance with institutional and international ethical standards.

Study setting and participant recruitment

Participant flow through the study is illustrated in a CONSORT flow diagram (Fig. 1). A total of 45 patients with periodontitis were assessed for eligibility; 27 met the inclusion criteria and were enrolled. Eighteen patients were excluded due to not meeting eligibility criteria or incomplete follow-up. The study enrolled 27 patients, comprising 81 periodontal quadrants and 596 sites—212 in the DLT-H group, 189 in the DLT-L group, and 195 in the SRP group—with a mean age of 33.6 years. All randomized quadrants and sites completed follow-up and were included in the final analysis. Participants were recruited from both the male and female branches

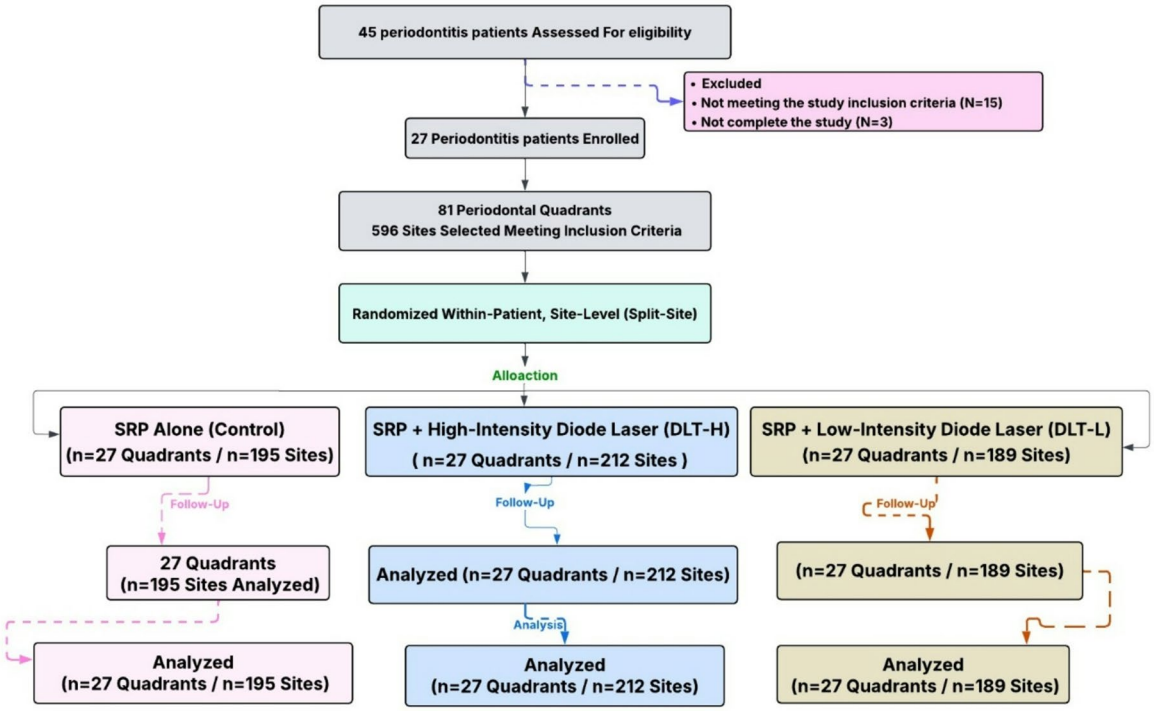


Fig. 1. CONSORT flow diagram of participants and periodontal sites through the split-site randomized clinical trial.

of the Dental Polyclinics, Faculty of Dentistry, UST, between October 2024 and February 2025. All eligible individuals were invited to participate voluntarily and received detailed verbal and written information about the study objectives, potential risks and benefits, and their rights as participants. Written informed consent was obtained before enrollment. Baseline demographic data, medical history, oral hygiene status, and findings from a comprehensive intraoral examination were recorded. To minimize inter-examiner variability, all clinical assessments and procedures were performed by a single calibrated examiner.

Eligibility criteria

Eligible participants were adults aged 25 to 58 years with a diagnosis of generalized periodontitis according to the 2018 classification, exhibiting PPD from 5 to 9 mm in at least 30% of sites, and possessing a minimum of 20 functional teeth, excluding third molars. Exclusion criteria encompassed a history of smoking or qat chewing, pregnancy or breastfeeding, and the use of systemic medications known to affect periodontal status. Patients with systemic conditions such as diabetes mellitus, autoimmune disorders, or hematological diseases that could influence periodontal inflammation or healing were also excluded. Furthermore, individuals who had received periodontal therapy or systemic antibiotics within the preceding three months, or who had fixed or removable prosthetic appliances or orthodontic devices in the treatment regions, were not eligible for participation.

Sample size calculation

The sample size was calculated based on interleukin-1β (IL-1β) concentration as the primary outcome, using the approach described by Talmac et al. (2019)[20]. IL-1β was selected for power estimation due to its higher expected variability, thereby providing a conservative calculation applicable to both immunological and clinical endpoints. Furthermore, the final sample size of 27 patients exceeds the minimum calculated requirement and is comparable to, or slightly larger than, previously published randomized clinical trials evaluating adjunctive diode laser therapy in periodontitis, supporting the adequacy of the study design[21–23].

With a standard deviation of 38.3 pg/mL, an expected effect size of 33.51 pg/mL, a two-sided alpha of 0.05, and 80% statistical power, the minimum required sample was 21 sites per group. To account for potential dropouts and maintain sufficient power, a total of 27 patients—each contributing three non-adjacent periodontal sites (totaling 81 sites)—were enrolled and randomly assigned to the three treatment groups.

A standardized data collection form was employed to capture each participant's demographic information, medical history, and clinical periodontal measurements at every visit. All assessments were conducted on a dental chair under standard lighting conditions using a dedicated dental examination kit.

Primary and secondary outcomes

The primary outcome of the study was the change in clinical attachment level (CAL) at 3 months (T2). Secondary outcomes included changes in probing pocket depth (PPD), gingival index (GI), and plaque index (PI) at 1 month (T1) and 3 months (T2).

Randomization, blinding, and treatment allocation

All periodontal sites were initially examined, and those presenting probing depth ≥ 5 mm, no furcation involvement, and no extensive restorations were considered eligible. From these eligible sites, three non-adjacent sites were selected for inclusion in the trial to represent the three treatment modalities. Owing to the nature of laser therapy, blinding of the operator was not feasible, and no sham laser was employed.

Within-patient randomization (site-level allocation)

For each enrolled patient, three non-adjacent periodontal sites were identified and designated as Site A, Site B, and Site C. The study followed a within-patient, split-site randomized design. A random allocation sequence assigning the three treatment codes (1 = SRP, 2 = SRP + DLT-H, 3 = SRP + DLT-L) to these sites was generated individually for each patient as a permutation of the three labels. The complete set of permutations was created centrally by an independent statistician using a computer-generated random number sequence to ensure equal probability of assignment and prevent selection bias. Each patient's allocation was placed in a sequentially numbered, opaque, sealed envelope, which was opened only after baseline measurements and site selection were completed. Outcome assessment was performed by a blinded examiner who was unaware of the treatment allocation. Patients were not informed of the specific treatment modality applied to each site.

Following an initial screening that comprised a comprehensive medical and dental history along with a thorough intraoral evaluation, 27 patients met the study's inclusion and exclusion criteria. These participants contributed a total of 81 periodontal quadrants encompassing 596 sites: 212 sites allocated to the DLT-H group, 189 to the DLT-L group, and 195 to the SRP group, all diagnosed with periodontitis.

The subjects were then randomly divided into three groups:

- Group I (Control): SRP alone.
- Group II (Test 1): SRP + high-intensity diode laser (DLT-H).
- Group III (Test 2): SRP + low-intensity diode laser (DLT-L).

Both participants and the outcome assessor were blinded to group allocation. All procedures were standardized and performed by the same periodontist to minimize variability.

Experimental design

Prior therapy, all participants received standardized oral hygiene instructions, including the Bass brushing technique using a soft-bristled toothbrush and fluoridated toothpaste (Colgate® Cavity Protection), ensuring uniformity by using the same toothpaste for all. Instructions emphasized brushing twice daily for at least three minutes and proper interdental cleaning. The use of chemical mouth rinses was prohibited to prevent confounding effects. Participants were continuously motivated to maintain effective plaque control throughout the study. Full mouth supragingival debridement was performed using ultrasonic scalers, followed by polishing with abrasive paste and natural bristle brushes. Subgingival SRP were then carried out with Gracey curettes under local anesthesia as needed.

For the test groups, adjunctive diode laser therapy was applied during the same appointment immediately following SRP, using a 980 nm DiodeLXSMART laser (LASOTRONIX, Poland), delivered via flexible optical fibers (SMA 905 connector, 320 µm core diameter). Based on the fiber core area (0.000804 cm^2), the calculated power densities were 2.49×10^3 W/cm^2 for the high-intensity setting (2 W) and 6.22×10^2 W/cm^2 for the low-intensity setting (0.5 W). These settings were selected in accordance with previously published clinical protocols[19]. Correspondingly, the energy densities were 2.49×10^4 J/cm^2 for the high-intensity protocol (2 W, 10-s exposure) and 1.87×10^4 J/cm^2 for the low-intensity protocol (0.5 W, 30-s exposure) as in Suppl. 1. For DLT-L, the laser was applied in a non-contact, defocused mode at a short distance from the tissue surface. Under these conditions, beam divergence results in an effective irradiation field larger than the fiber tip diameter; therefore, the actual power density delivered to the tissue is lower than the nominal value calculated based on tip dimensions. Moreover, irradiation was applied dynamically using a sweeping motion within the periodontal pocket, leading to energy distribution across the entire treated surface. Therefore, the effective fluence at any single tissue point is substantially lower than the nominal value calculated from fiber dimensions. The selected parameters were based on previously published clinical protocols and were intended to provide anti-inflammatory photobiomodulation under clinical conditions[19].

In accordance with the manufacturer's safety specifications, the device includes an automatic thermal-protection feature that interrupts laser emission for 30 s whenever continuous irradiation exceeds 10 s. Consequently, all laser applications in both the DLT-H and DLT-L groups were delivered in cycles not exceeding 10 s per continuous exposure to ensure safe and standardized energy delivery across all treated sites.

The optical fiber was carefully introduced into each periodontal pocket, after which the laser was activated. While the laser emitted light, the fiber was gently advanced from the apical toward the coronal region in a controlled sweeping motion to achieve even energy distribution and limit heat buildup. This procedure was systematically applied to the mesial, distal, buccal, and lingual aspects of each site and repeated until the entire pocket had been fully irradiated. To prevent thermal injury, the device automatically paused laser emission for 30 s whenever the irradiation time exceeded 10 s. All interventions were carried out under local anesthesia, and both the patients and the operator wore appropriate protective eyewear[24].

- *DLT-H Group*: High-intensity settings (2.0 W, continuous mode, in contact mode) for decontamination, coagulation, and epithelial debridement.
- *DLT-L Group*: Low-intensity settings (0.5 W, continuous mode, in non-contact mode) for photobiomodulation to stimulate cellular repair and modulate inflammation.

No adjunctive systemic antibiotics or antiseptics were prescribed.

Clinical assessments were performed at baseline (T0), 1 month (T1), and 3 months (T2), including:

1. *Plaque Index (PI)* – Silness and Löe (1964)[1], scored 0–3 across four tooth surfaces; mean PI calculated per tooth and per participant.
2. *Gingival Index (GI)* – Löe and Silness (1963)[2], scored 0–3 across four surfaces; mean GI calculated per tooth and per participant.
3. *Probing Pocket Depth (PPD)* – Distance from gingival margin to pocket base, measured at six sites per tooth using a UNC-15 periodontal probe.
4. *Clinical Attachment Level (CAL)* – Distance from cementoenamel junction to pocket base, measured at six sites per tooth.

Measurements for each index tooth were taken from four surfaces and averaged per tooth. To account for clustering of multiple sites within the same patient, these tooth-level means were further averaged for each corresponding quadrant; quadrant-level means were then used for statistical analyses.

Prior to the evaluation process, the examiner (AAA) underwent structured calibration training to ensure consistency, accuracy, and adherence to the assessment criteria. To ensure consistency and accuracy, all clinical assessments were conducted by a single experienced examiner (AAA). Oversight was provided by a senior periodontist (WAH) with over two decades of clinical experience, who actively supervised the procedures and contributed to the evaluation of periodontal parameters. Their guidance helped maintain high reliability and uniform interpretation of the data. Measurements for each index tooth were taken from four surfaces, averaged per tooth, and then further averaged for each corresponding quadrant. The examiner was calibrated prior to data collection, achieving excellent intra-examiner reliability (Kappa = 0.90), which reinforced the credibility and reproducibility of the recorded clinical indices.

Statistical analysis

Data were recorded in Excel, verified, coded, and analyzed using SPSS v25.0. Continuous variables were expressed as mean ± SD, and categorical variables as frequency and percentage. Normality was assessed before analysis. Within-group changes across T0, T1, and T2 were evaluated using repeated-measures ANOVA or the Friedman test. If significant, post hoc pairwise comparisons were performed as appropriate. Between-group differences (SRP, DLT-H, DLT-L) were analyzed using one-way ANOVA or the Kruskal-Wallis test. If significant, post hoc comparisons were used as appropriate. The statistical significance level was set at $p < 0.05$. Bland–Altman analysis was used to evaluate agreement between the two treatment methods and to identify any systematic bias, ensuring reliability of the results.

Results

The study included 27 participants (19 females, 70.4%; 8 males, 29.6%) with a mean age of 33.6 years. A total of 81 periodontal quadrants were evenly allocated to three treatment modalities—SRP, scaling with DLT-H, and scaling with DLT-L—each representing 33.3% of sites. Sites were evenly distributed across quadrants: 22.2% upper right, 19.8% upper left, 27.1% lower right, and 30.9% lower left (Table 1).

CAL gain was 2.20 ± 0.45 mm in the DLT-H group compared with 1.72 ± 0.51 mm in the SRP group ($p = 0.001$).

Clinical assessments demonstrated significant, time-dependent improvements across all periodontal parameters. At one-month post-treatment, all groups showed reductions in plaque accumulation and gingival inflammation, along with decreased PPD and gains in clinical attachment, which were further enhanced at the three-month follow-up (Suppl. 2).

All treatment modalities produced statistically significant improvements over time ($p < 0.001$). In the SRP group, mean PI decreased from 1.83 ± 0.35 at baseline to 0.93 ± 0.22 at one month and 0.51 ± 0.16 at three

Parameter	Category	*N*	%	*P*
Gender	Female	19	70.4	< 0.001
	Male	8	29.6	
Oral cavity side	Upper right	18	22.2	0.492
	Upper left	16	19.8	
	Lower right	22	27.1	
	Lower left	25	30.9	
	Treatment type	N	Mean ± SD	
Number of treated teeth	SRP	104	3.85 ± 0.77	0.281
	DLT-H	108	3.96 ± 0.71	
	DLT-L	97	3.67 ± 0.56	

Table 1. Gender distribution, treatment type, and treatment mouth side of the periodontitis patients under study. *SRP* Scaling and root planning, *DLT-L* low-intensity diode laser treatment, *DLT-H* high-intensity diode laser treatment.

Periodontal measurements	Time of treatment			*P*-value
	T0	T1	T2	
SRP				
PI	1.83 ± 0.35	0.93 ± 0.22	0.51 ± 0.16	< 0.001
GI	1.97 ± 0.31	0.98 ± 0.25	0.57 ± 0.16	< 0.001
PPD	3.94 ± 0.38	2.98 ± 0.31	2.74 ± 0.20	< 0.001
CAL	5.16 ± 0.74	4.18 ± 0.63	3.43 ± 0.53	< 0.001
DLT-H				
PI	1.86 ± 0.35	0.84 ± 0.25	0.40 ± 0.17	< 0.001
GI	2.01 ± 0.30	0.88 ± 0.25	0.43 ± 0.15	< 0.001
PPD	3.96 ± 0.51	2.89 ± 0.29	2.64 ± 0.18	< 0.001
CAL	5.19 ± 0.80	3.95 ± 0.67	2.99 ± 0.60	< 0.001
DLT-L				
PI	1.79 ± 0.34	0.87 ± 0.25	0.50 ± 0.17	< 0.001
GI	1.94 ± 0.27	0.90 ± 0.21	0.51 ± 0.15	< 0.001
PPD	3.86 ± 0.39	2.91 ± 0.28	2.68 ± 0.22	< 0.001
CAL	5.04 ± 0.75	4.01 ± 0.63	3.20 ± 0.58	< 0.001

Table 2. Effectiveness of different types of periodontal treatment in patients with periodontitis at different times. *PI* Plaque Index, *GI* Gingival Index, *PPD* Probing Pocket Depth, *CAL* Clinical Attachment Level, *SRP* scaling and root planning, *DLT-H* high-intensity diode laser treatment, *DLT-L* Low-Intensity diode laser treatment, *T0* Base line, *T1* Treatment after one month, *T2* Treatment after three months.

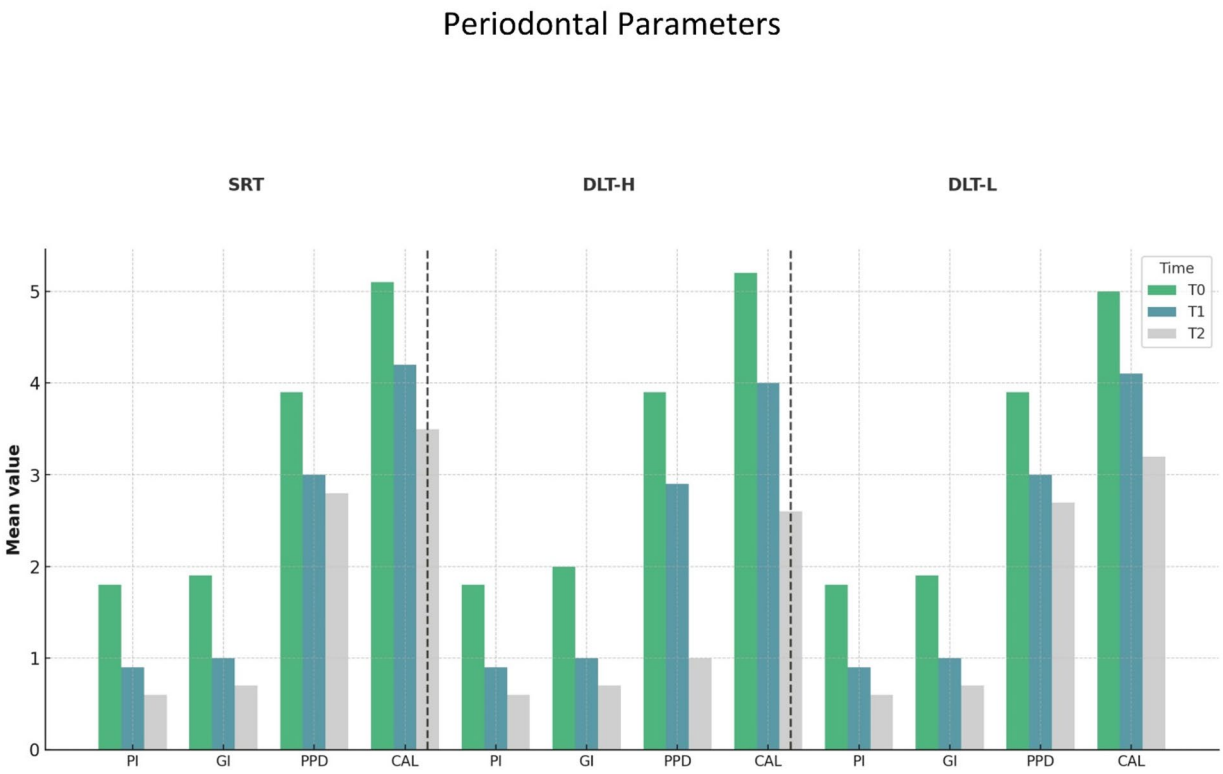


Fig. 2. Mean Periodontal Parameter Changes (PI, GI, PPD, CAL) Over Time (T0, T1, T2) Across Treatment Groups (SRP, DLT-H, DLT-L).

months, while GI declined from 1.97 ± 0.31 to 0.98 ± 0.25 and 0.57 ± 0.16. PPD and CAL improved progressively (PPD: 3.94 ± 0.38 mm to 2.74 ± 0.20 mm; CAL: 5.16 ± 0.74 mm to 3.43 ± 0.53 mm). Both DLT-H and DLT-L groups exhibited greater improvements, with DLT-H achieving the largest gains (PI: 1.86 ± 0.35 to 0.40 ± 0.17; GI: 2.01 ± 0.30 to 0.43 ± 0.15; PPD: 3.96 ± 0.51 mm to 2.64 ± 0.18 mm; CAL: 5.19 ± 0.80 mm to 2.99 ± 0.60 mm), while DLT-L also showed significant but slightly less pronounced improvements (CAL: 5.04 ± 0.75 mm to 3.20 ± 0.58 mm) as shown in Table 2; Fig. 2.

Parameter	Type of treatment			*P*-value
	SRP	DLT-H	DLT-L	
Δ (T0-T1)				
PI	0.90 ± 0.28	1.02 ± 0.31	0.92 ± 0.29	0.241
GI	0.99 ± 0.30	1.13 ± 0.33	1.03 ± 0.25	0.148
PPD	0.96 ± 0.47	1.07 ± 0.46	0.96 ± 0.44	0.499
CAL	0.98 ± 0.39	1.24 ± 0.33	1.03 ± 0.35	0.001*
Δ (T0-T2)				
PI ά	1.32 ± 0.34	1.46 ± 0.28	1.30 ± 0.29	0.103
GI	1.40 ± 0.26	1.58 ± 0.32	1.43 ± 0.24	0.020*
PPD ά	1.20 ± 0.39	1.32 ± 0.48	1.18 ± 0.30	0.365
CAL ά	1.72 ± 0.51	2.20 ± 0.45	1.85 ± 0.39	0.001*
Δ (T1-T2)				
PI	0.42 ± 0.22	0.44 ± 0.17	0.38 ± 0.19	0.228
GI	0.41 ± 0.16	0.45 ± 0.14	0.40 ± 0.15	0.368
PPD	0.24 ± 0.26	0.25 ± 0.20	0.23 ± 0.27	0.842
CAL	0.74 ± 0.20	0.96 ± 0.32	0.82 ± 0.31	0.005*

Table 3. Mean and standard deviation of the four basic clinical periodontal parameters assessed by different types of treatment. *PI* Plaque Index, *GI* Gingival Index, *PPD* Probing Pocket Depth, *CAL* Clinical Attachment Level, *DLT-H* High-intensity diode laser treatment, *DLT-L* Low intensity diode laser treatment, *SRP* Scaling and Root Planning treatment, *T0* Base line, *T1* Treatment after one month, *T2* Treatment after three months, asterisks (*): for significance., asterisks (ά): for normality distribution.

Δ (T0-T1)						
Parameter	Treatment		MD	95% CI for Difference		P
				Lower	Upper	
PI	SRP	DLT-H	−0.12	−0.32	0.07	0.380
		DLT-L	−0.02	−0.22	0.17	1.000
	DLT-H	DLT-L	0.10	−0.10	0.29	0.645
GI	SRP	DLT-H	−0.14	−0.34	0.06	0.262
		DLT-L	−0.04	−0.24	0.16	1.000
	DLT-H	DLT-L	0.10	−0.10	0.29	0.674
PPD	SRP	DLT-H	−0.12	−0.42	0.19	1.000
		DLT-L	0.00	−0.31	0.30	1.000
	DLT-H	DLT-L	0.11	−0.19	0.42	1.000
CAL	SRP	DLT-H	−0.26	−0.50	−0.02	0.001*
		DLT-L	−0.05	−0.29	0.19	0.796
	DLT-H	DLT-L	0.21	−0.03	0.45	0.037*

Table 4. Pairwise comparisons between treatment types for each clinical parameter for change from pre-treatment point 1 (T0) compared to 1 month treatment (T1). *PI* Plaque Index, *GI* Gingival Index, *PPD* Probing Pocket Depth, *CAL* Clinical Attachment Level, *DLT-H* High-intensity diode laser treatment, *DLT-L* Low intensity diode laser treatment, *MD* mean difference, *CI* confidence interval, *T0* Base line, *T1* Treatment after one month, asterisks (*): for significance.

Within-group comparisons across baseline, one-month, and three-month intervals were all statistically significant ($p < 0.001$). The DLT-H group consistently demonstrated the greatest reductions in PI, GI, PPD, and the highest CAL gains, followed by DLT-L and then SRP (Suppl. 3).

Comparative analysis revealed that during the short-term interval (T0–T1), all groups achieved similar reductions in PI, GI, and PPD, while DLT-H showed significantly greater CAL gain ($p = 0.001$). Over the three-month period (T0–T2), DLT-H exhibited the most pronounced improvements across all parameters, particularly CAL (2.20 ± 0.45 mm) compared with SRP (1.72 ± 0.51 mm) and DLT-L (1.85 ± 0.39 mm), with significant differences in GI ($p = 0.020$) and CAL ($p = 0.001$). Between one and three months (T1–T2), intergroup differences in PI, GI, and PPD were not significant, whereas CAL remained significantly higher in the DLT-H group ($p = 0.005$) as shown in Tables 3, 4, 5 and 6.

Bland–Altman analysis was applied to evaluate agreement among treatment modalities. The plots indicated stable variation patterns, with DLT-H showing comparatively narrower limits of agreement across time

Δ (T0-T2)						
Parameter	Treatment		MD	95% CI for Difference		P
				Lower	Upper	
PI	SRP	DLT-H	−0.15	−0.35	0.06	0.245
		DLT-L	0.02	−0.19	0.22	1.000
	DLT-H	DLT-L	0.16	−0.04	0.37	0.157
GI	SRP	DLT-H	−0.18	−0.36	0.00	0.020*
		DLT-L	−0.03	−0.21	0.15	1.000
	DLT-H	DLT-L	0.15	−0.03	0.33	0.141
PPD	SRP	DLT-H	−0.13	−0.39	0.14	0.742
		DLT-L	0.01	−0.25	0.28	1.000
	DLT-H	DLT-L	0.14	−0.12	0.40	0.592
CAL	SRP	DLT-H	−0.48	−0.78	−0.18	< 0.001*
		DLT-L	−0.12	−0.42	0.18	0.335
	DLT-H	DLT-L	0.36	0.06	0.66	0.018

Table 5. Pairwise comparisons between the treatment types for each clinical parameter for the change from pre-treatment point 1 (T0) compared to 3-month treatment (T2). *PI* Plaque Index, *GI* Gingival Index, *PPD* Probing Pocket Depth, *CAL* Clinical Attachment Level, *DLT-H* High-intensity diode laser treatment, *DLT-L* Low intensity diode laser treatment, *MD* mean difference, *CI* confidence interval, *T0* Base line, *T2* Treatment after three months, asterisks (*): for significance.

Δ (T1-T2)						
Parameter	Treatment		MD	95% CI for Difference		P
				Lower	Upper	
PI	SRP	DLT-H	−0.02	−0.15	0.10	1.000
		DLT-L	0.04	−0.09	0.17	1.000
	DLT-H	DLT-L	0.06	−0.06	0.19	0.670
GI	SRP	DLT-H	−0.04	−0.14	0.06	1.000
		DLT-L	0.01	−0.09	0.11	1.000
	DLT-H	DLT-L	0.05	−0.05	0.15	0.701
PPD	SRP	DLT-H	−0.01	−0.17	0.15	1.000
		DLT-L	0.02	−0.15	0.18	1.000
	DLT-H	DLT-L	0.03	−0.14	0.19	1.000
CAL	SRP	DLT-H	−0.22	−0.41	−0.03	0.004*
		DLT-L	−0.07	−0.26	0.12	0.932
	DLT-H	DLT-L	0.15	−0.04	0.34	0.091

Table 6. Pairwise comparisons between the treatment types for each clinical parameter from one month treatment (T1) compared to 3-month treatment (T2). *PI* Plaque Index, *GI* Gingival Index = GI, *PPD* Probing Pocket Depth, *CAL* Clinical Attachment Level, *DLT-H* High-intensity diode laser treatment, *DLT-L* Low-intensity diode laser treatment, *MD* mean difference, *CI* confidence interval, *T1* Treatment after one month, *T2* Treatment after three months, asterisks (*): for significance.

points. As the method does not assess treatment superiority, these findings were considered descriptive rather than indicative of clinical benefit. Accordingly, treatment efficacy was determined solely through primary statistical analyses, which demonstrated that DLT-H produced the greatest improvements in CAL and gingival inflammation, whereas PI and PPD changes were modest. DLT-L showed moderate, delayed responses, while SRP alone yielded the smallest overall improvements as shown in Table 7; Fig. 3.

Discussion

SRP remains the cornerstone of nonsurgical periodontal therapy. However, the mechanical approach alone often faces limitations in completely eliminating bacterial biofilm and endotoxins, particularly in deep pockets and furcation areas, thereby justifying the use of adjunctive therapies. Among these, diode lasers have attracted increasing attention for their dual ability to exert antimicrobial effects and modulate host inflammatory responses[3]. Evidence suggests that diode lasers, depending on their energy settings, can facilitate bacterial

	PI		GI		PPD		CAL	
	MD	*P*-value	MD	*P*-value	MD	*P*-value	MD	*P*-value
Δ(T0-T1)								
SRP & DLH	−0.12	0.015*	−0.14	< 0.001*	−0.12	0.062	−0.26	0.003*
SRP& DLL	−0.02	0.553	−0.04	0.312	0.00	0.955	−0.05	0.560
DLH & DLL	0.10	0.009*	0.10	0.027*	0.11	0.087	0.21	0.010*
Δ(T0-T2)								
SRP & DLH	−0.15	0.007*	−0.18	< 0.001*	−0.13	0.096	−0.48	< 0.001*
SRP & DLL	0.02	0.703	−0.03	0.163	0.01	0.804	−0.12	0.244
DLH & DLL	0.16	0.001*	0.15	< 0.001*	0.14	0.037*	0.36	< 0.001*
Δ(T1-T2)								
SRP & DLH	−0.02	0.526	−0.04	0.057*	−0.01	0.861	−0.22	0.001*
SRP & DLL	0.04	0.224	0.01	0.763	0.02	0.691	−0.07	0.334
DLH & DLL	0.06	0.003*	0.05	0.061	0.03	0.654	0.15	0.061

Table 7. Bland–Altman mean differences (MD) and P-values Comparing Clinical Parameters (PI, GI, PPD, CAL) Between Treatment Modalities at Different Time Intervals (T0–T1, T0–T2, T1–T2). *PI* Plaque Index, *GI* Gingival Index, *PPD* Probing Pocket Depth, *CAL* Clinical Attachment Level, *DLT-H* High-intensity diode laser treatment, *DLT-L* Low intensity diode laser treatment, *MD* mean difference, *T0* Base line, *T1* Treatment after one month, *T2* Treatment after three months, asterisks (*): for significance.

reduction, enhance wound healing, and promote tissue repair and healing processes through collagen synthesis and angiogenesis[4,5].

The null hypothesis stated that no differences would exist in PI, GI, PPD, or CAL among the three treatment groups. Although all modalities produced significant within-group improvements over time ($p < 0.001$), inter-group analyses identified clear distinctions. Adjunctive DLT-H yielded significantly greater reductions in GI and greater CAL gain compared with both SRP alone and DLT-L (GI, $p = 0.020$; CAL, $p = 0.001$), with CAL superiority persisting at three months ($p = 0.004$). By contrast, between-group differences in PI and PPD were not significant. Consequently, the null hypothesis is rejected for GI and CAL but retained for PI and PPD, underscoring the enhanced periodontal benefits of high-intensity diode laser adjunctive treatment.

The 980 nm diode laser used in this study provides limited penetration, ensuring safe superficial ablation, while its effects differ from other lasers such as Nd: YAG, which interacts differently with tissues. In the present trial, all treatment groups (SRP, SRP + DLT-H, and SRP + DLT-L) demonstrated significant improvements in PI, GI, PPD, and CAL over the three-month period. The general improvements in PI and GI reflect effective biofilm disruption and improved oral hygiene, findings consistent with those of Pooja et al. (2019) and Cobb et al. (2002)[6,7]. The comparable outcomes can be explained by the similarity in diode laser specifications and application protocols.

Similarly, the reduction in PPD accompanied by CAL gains supports the effectiveness of nonsurgical approaches, with or without adjunctive lasers, as also reported by Heitz -Mayfield and Lang (2013)[8].

When comparing modalities, SRP alone produced significant early improvements that stabilized by the first month, which aligns with the typical healing line described in a systematic review by Needleman et al. (2005)[9] and corroborated by recent systematic evaluations[24,25]. These observations highlight the inherent limitations of SRP in achieving deeper and more sustained clinical improvements supporting the rationale for adjunctive approaches.

Adjunctive use of DLT-H resulted in both rapid and sustained improvements across all clinical parameters, with superior CAL gains persisting through three months. These findings are in line with previous investigations by Aoki et al. (2015)[26], Qadri et al. (2010)[27], Üstün et al. (2014)[28] and Abduljabbar et al. (2017)[29] which demonstrated greater periodontal healing with diode laser adjuncts compared to SRP alone, which may attributed to use of controlled fiber placement, standardized motion within the periodontal pocket, and precise energy delivery. The enhanced effects are likely attributable to a combination of bacterial reduction, de-epithelialization, and host modulation through fibroblast activation and extracellular matrix remodeling.

DLT-L also achieved significant improvements, particularly in PI and GI during the early phase, consistent with the photobiomodulatory effects reported by Dukić et al. (2012) and Gündoğar et al. (2016)[21,30]. Sustained but moderate improvements in CAL and GI at three months further support its role in modulating inflammation and promoting soft tissue repair, as documented by Qadri et al. (2005) and Saglam et al. (2014)[23,31]. The comparable outcomes are likely due to the shared fiber positioning and movement strategy, which ensures consistent energy delivery within the periodontal pocket. Nonetheless, conflicting findings exist, with some studies reporting negligible added benefit of LLLT over SRP[32,33] may be due to variations in laser parameters, pulse settings, methodology and timing of application, suggesting that patient characteristics and protocol variability may influence outcomes.

Direct comparisons revealed that DLT-H consistently outperformed both SRP and DLT-L, particularly in CAL improvement. These results are consistent with Kreisler et al. (2005)[22], who demonstrated superior periodontal clinical outcomes with high-intensity laser therapy. Similar outcomes were observed by Moritz et al. (1997) and Lui et al. (2011)[34,35], where diode lasers achieved greater PD reduction and decreased gingival inflammation

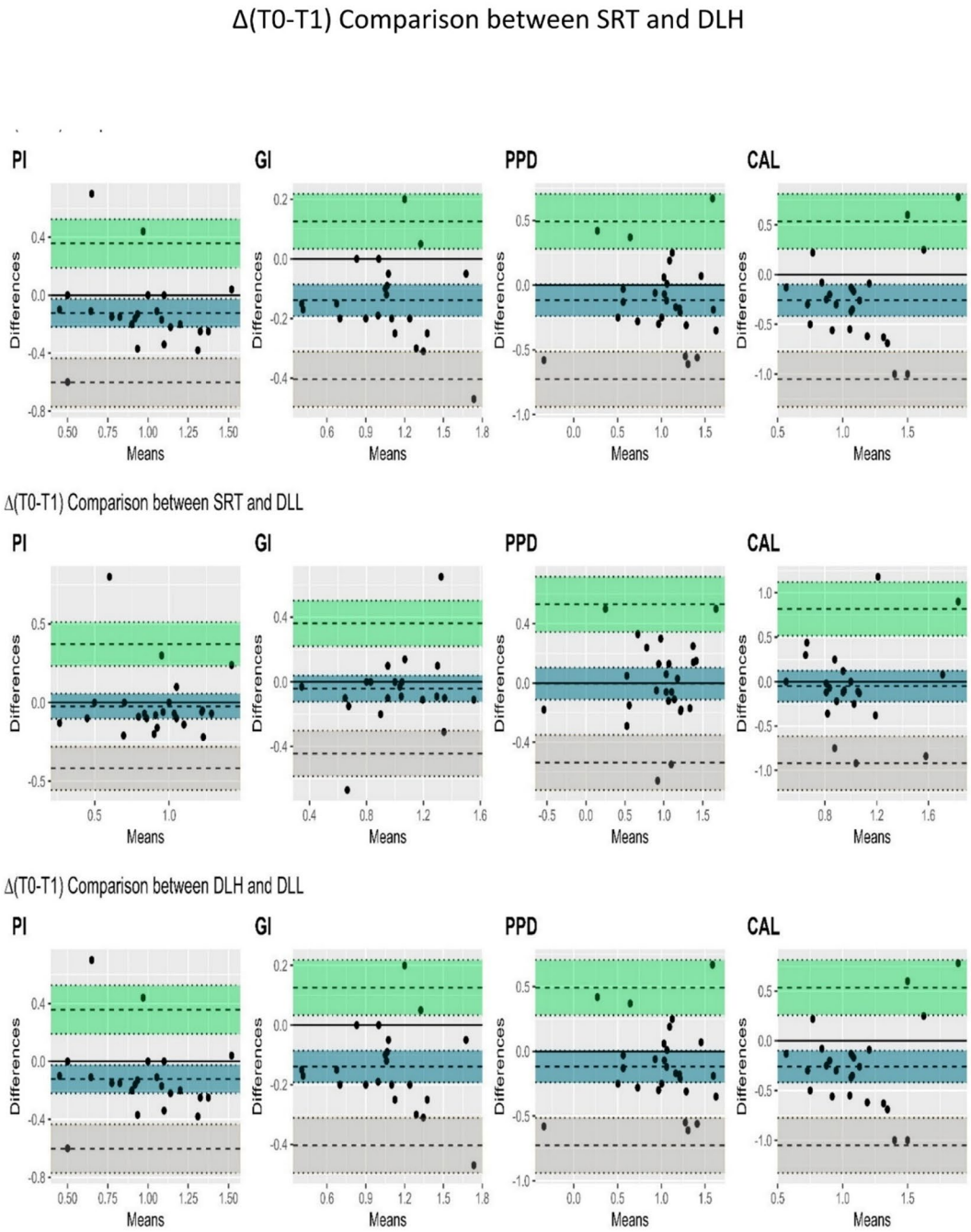


Fig. 3. Bland–Altman Plots Showing Δ(T0–T1) Differences in Periodontal Parameters (PI, GI, PPD, CAL) Among Treatment Comparisons: SRT vs. DLT-H, SRT vs. DLT-L, and DLT-H vs. DLT-L.

compared with SRP alone. On the other hand, some trials, including Pangam (2018)[36], Yilmaz et al. (2002)[37], and De Micheli et al. (2011)[38], reported no significant additional benefit of diode lasers over SRP, highlighting variability in treatment efficacy across studies, which may be attributed to variations in laser wavelength, output power, fiber diameter, irradiation timing, and procedural methodology, including the number of sessions.

While direct comparisons in this study confirmed the superior CAL improvement with DLT-H, it is important to acknowledge that not all previous trials have reported additional benefits of diode lasers over SRP. Variability in outcomes across studies may be explained by differences in laser wavelength, output power, fiber diameter, irradiation technique, number of sessions, and patient characteristics. By considering these factors, these findings align with the subset of literature demonstrating enhanced periodontal improvement with high-intensity diode lasers, while highlighting the potential influence of protocol-specific parameters on treatment efficacy. Regarding to PI, PI reductions likely reflect enhanced patient oral hygiene rather than a direct effect of laser therapy; thus, conclusions focus on short-term CAL and GI improvements Although they were statistically significant, the observed differences in (CAL across groups were modest. Given the inherent heterogeneity

of clinical assessments and the brief follow-up time, these results should be taken cautiously since statistical significance may not always translate into a clinically relevant effect.

In the late healing phase (T1–T2), CAL gains remained significantly higher in the DLT-H group, whereas PI, GI, and PPD stabilized across groups. This sustained advantage of DLT-H aligns with Saglam et al. (2014) and Matarese et al. (2017)[31,39], who emphasized its clinical attachment outcomes potential. Conversely, the DLT-L group showed moderate but less durable improvements, possibly due to lower penetration depth and reduced biostimulatory capacity, as suggested by Ren et al. (2017)[33].

While DLT-L demonstrated measurable clinical improvements, these effects were consistently lower than those achieved with DLT-H. This indicates that using DLT-L alone may not provide optimal therapeutic outcomes. The findings support the notion that high-intensity diode laser application remains the most effective approach for promoting clinical attachment gain and reducing gingival inflammation, whereas low-intensity settings may serve better as an adjunct to high-intensity therapy or for gentle biostimulation rather than as a standalone treatment. Safety was monitored throughout the study by assessing postoperative pain, soft tissue burns, ulceration, or necrosis. Additionally, no laser-related adverse events were observed.

Collectively, these findings confirm that while all treatment modalities yielded clinically meaningful improvements, the magnitude and persistence of benefits were greatest with DLT-H, followed by DLT-L, with SRP alone showing the least durable outcomes. This evidence supports the adjunctive use of diode lasers, particularly at high intensity, as a promising strategy to enhance periodontal healing responses and sustain periodontal health beyond conventional mechanical therapy.

Clinical implications

The findings of this study indicate that adjunctive diode laser therapy enhances the outcomes of nonsurgical periodontal treatment. DLT-H produced the most pronounced reductions in gingival inflammation and the greatest gains in CAL, surpassing both SRP alone and DLT-L. Although both laser modalities contributed to soft-tissue healing, DLT-H provided the most consistent and durable improvements.

These results carry several important clinical implications. Diode laser therapy enhances periodontal healing and microbial control, particularly in deep or anatomically complex sites that are challenging for conventional mechanical debridement. The adjunctive use of lasers may also reduce the need for systemic antimicrobials, supporting more conservative, antibiotic-sparing approaches. Additionally, accelerated resolution of inflammation and improved patient comfort can increase treatment acceptance and adherence.

From a treatment-planning perspective, high-intensity diode laser is applied to sites with moderate-to-deep pockets or persistent inflammation, based on the clinical scenario and tissue pathology. Low-intensity diode laser may be considered as an adjunctive modality for maintenance therapy through photobiomodulation, exerting dose-dependent anti-inflammatory and tissue-modulating effect. When applied appropriately by trained clinicians, this tailored approach supports optimal soft tissue healing and enhances the magnitude and durability of clinical outcomes.

The findings of the present study indicate that adjunctive DLT-H therapy provides greater short-term improvements in clinical parameters, particularly clinical CAL and gingival inflammation, compared with DLT-L and SRP alone. These results support the potential benefit of DLT-H as an adjunct to SRP. However, no microbiological, biochemical, or long-term assessments were conducted; therefore, any implications regarding microbial control, antibiotic use, or specific recommendations for pocket depth should be interpreted cautiously. Further studies with extended follow-up and paraclinical evaluations are warranted to substantiate broader clinical applications. In terms of clinical relevance, DLT-H resulted in greater CAL gains than DLT-L (2.20 mm vs. 1.85 mm), the difference (~0.35 mm) may be modest in clinical terms. These results indicate short-term attachment improvement rather than definitive long-term periodontal regeneration.

This study evaluated clinical outcomes over a three-month period. While DLT-H showed the greatest short-term improvements in CAL and GI, the long-term stability of these gains remains unknown, as periodontal therapy typically requires 6–12 months of follow-up to assess durable effects. The relatively small sample size, single-center design, and potential variability in host response and operator technique may limit the generalizability of the findings. Future well-designed trials with extended follow-up and larger cohorts are warranted to confirm the persistence and broader applicability of these results.

Additionally, site-level randomization was applied, analyses used quadrant-level averages and did not formally account for within-patient correlation. This may influence the precision of estimated effects and should be considered a limitation of the study.

Conclusions

This study demonstrates that while all nonsurgical modalities improved periodontal parameters, DLT-H adjunctive therapy provided the greatest and most sustained clinical benefits, particularly in clinical attachment level. DLT-L offered moderate improvements, whereas SRP alone showed initial gains that stabilized over time. These findings support the selective use of DLT-H to enhance clinical improvement, with further studies needed to confirm long-term outcomes.

Data availability

All data generated or analyzed during this study are included in this published article.

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References

1. Silness, J. & Löe, H. Periodontal disease in pregnancy part II. Correlation between oral hygiene and periodontal condition. *Acta Odontol. Scand.* **22**(1), 121–135 (1964).
2. Löe, H. & Silness, J. Periodontal disease in pregnancy part I. Prevalence and severity. *Acta Odontol. Scand.* **21**(6), 533–551 (1963).
3. Isola, G. et al. New frontiers on adjuvants drug strategies and treatments in periodontitis. *Sci. Pharm.* **89** (4), 46 (2021).
4. Aoki, A., Sasaki, K. M., Watanabe, H. & Ishikawa, I. Lasers in nonsurgical periodontal therapy. *Periodontol. 2000* **36**(1), 59–97 (2004).
5. Harris, D. M. & Yessik, M. Therapeutic ratio quantifies laser antisepsis: Ablation of *Porphyromonas gingivalis* with dental lasers. *Lasers Surg. Med.* **35**(3), 206–13 (2004).
6. Cobb, C. M. Clinical significance of non-surgical periodontal therapy: An evidence-based perspective of scaling and root planing. *J. Clin. Periodontol.* **29**, 22–32 (2002).
7. Pooja, M., Kashyap, R. S., Hegde, S., Kumar, M. A. & Boloor, V. Comparison of effectiveness of diode laser as an adjunct to scaling and root planing and scaling, root planing alone in treatment of chronic periodontitis—A randomized split-mouth clinical study. *J. Contemp. Dent.* **20199**(3), 117–23 (2019).
8. Heitz-Mayfield, L. J. & Lang, N. P. Surgical and nonsurgical periodontal therapy. Learned and unlearned concepts. *Periodontol. 2000* **62**(1), 218–31 (2013).
9. Needleman, I., Suvan, J., Moles, D. R. & Pimlott, J. A systematic review of professional mechanical plaque removal for prevention of periodontal diseases. *J. Clin. Periodontol.* **32**, 229–82 (2005).
10. Theodoro, L. H., Marcantonio, R. A. C., Wainwright, M. & Garcia, V. G. LASER in periodontal treatment: Is it an effective treatment or science fiction?. *Braz. Oral Res.* **35**, e099 (2021).
11. Sezen, D., Hatipoğlu, M. & Üstün, K. Evaluation of the clinical and biochemical efficacy of erbium, chromium: Ytrium-scandium-gallium-garnet (ER, CR: YSGG) laser treatment in periodontitis. *Lasers Med. Sci.* **35**(7), 1567–1575 (2020).
12. Al Shbool, B., Habashneh, R., Ta'ani, D. & Alzoubi, M. Clinical effectiveness of diode laser as an adjunct in the treatment of periodontitis. *Int. J. Oral Dent. Health*. **7**, 137 (2021).
13. Limeira, W. M. J. et al. The effectiveness of high-power diode laser as an adjunct to mechanical instrumentation of deep pockets in a patient with generalized, stage III, grade C periodontitis. *RGO* **72**, e20240006 (2024).
14. Park, J. S., Jung, H., Ryu, J.-J., Koo, K.-T. & Lee, J. Effect of erbium, chromium-doped: Yttrium, scandium, gallium, and garnet laser-assisted periodontal therapy using radial firing tip during early healing period: A randomized controlled split-mouth clinical trial. *BMC Oral Health* **24**(1), 539 (2024).
15. Sayar, F., Hashemi, S., Chiniforush, N., Seyed Jafari, E. & Jandaghi, A. Effects of diode and erbium lasers as an adjunct to scaling and root planing on clinical and immunological parameters in non-surgical periodontal treatment: A split-mouth randomized controlled clinical trial—effects of lasers on immunological parameters. *Lasers Med. Sci.* **37**(7), 3021–3030 (2022).
16. Soundarajan, S. & Rajasekar, A. Comparative evaluation of combined efficacy of methylene blue mediated antimicrobial photodynamic therapy (a-PDT) using 660 nm diode laser versus Erbium-chromium-yttrium-scandium-gallium-garnet (Er, Cr: YSGG) laser as an adjunct to scaling and root planing on clinical parameters in supportive periodontal therapy: A randomized split-mouth trial. *Photodiagn. Photodyn. Ther.* **39**, 102971 (2022).
17. Kulinska-Michalska, M., Cwiklinska, N., Lisiecka, M., Czaplicka-Szydlik, P. & Lewkowicz, N. The use of diode lasers in non-surgical therapy of periodontitis. *J. Pre-Clin. Clin. Res.* **14**(4), 169–173 (2020).
18. Katsikanis, F., Strakas, D. & Vouros, I. The application of antimicrobial photodynamic therapy (aPDT, 670 nm) and diode laser (940 nm) as adjunctive approach in the conventional cause-related treatment of chronic periodontal disease: A randomized controlled split-mouth clinical trial. *Clin. Oral Invest.* **24**(5), 1821–1827 (2020).
19. Badeia, R. A., Rajab, M. S. & Hazeem, M. I. Efficacy of diode laser with different intensities as an adjunct to traditional mechanical treatment of chronic periodontitis. *Mustansiria Dent. J.* **10**(1), 137–46 (2013).
20. Talmac, A. C., Calisir, M., Eroglu, E. G. & Ertugrul, A. S. Effects of Er, Cr: YSGG and Diode Lasers on Clinical Parameters and Gingival Crevicular Fluid IL-1β and IL-37 Levels in Generalized Aggressive Periodontitis. *Mediators Inflamm.* **2019**(1), 2780794 (2019).
21. Gündoğar, H., Şenyurt, S. Z., Erciyas, K., Yalım, M. & Üstün, K. The effect of low-level laser therapy on non-surgical periodontal treatment: A randomized controlled, single-blind, split-mouth clinical trial. *Lasers Med. Sci.* **31**(9), 1767–73 (2016).
22. Kreisler, M., Al Haj, H. & d'Hoedt, B. Clinical efficacy of semiconductor laser application as an adjunct to conventional scaling and root planing. *Lasers Surg. Med.* **37**(5), 350–355 (2005).
23. Qadri, T., Miranda, L., Tuner, J. & Gustafsson, A. The short-term effects of low-level lasers as adjunct therapy in the treatment of periodontal inflammation. *J. Clin. Periodontol.* **32** (7), 714–719 (2005).
24. Vinel, A. et al. Non-surgical periodontal treatment: SRP and innovative therapeutic approaches. In *Periodontitis: Advances in Experimental Research* 303–27 (Springer, 2022).
25. Calciolari, E. et al. The efficacy of adjunctive periodontal therapies during supportive periodontal care in patients with residual pockets. A systematic review and meta-analysis. *J. Periodontal Res.* **57**(4), 671–89 (2022).
26. Aoki, A. et al. <article-title update="added">Periodontal and peri-implant wound healing following laser therapy. *Periodontology 2000* **68**(1), 217–69 (2015).
27. Qadri, T., Poddani, P., Javed, F., Tunér, J. & Gustafsson, A. A short-term evaluation of Nd: YAG laser as an adjunct to scaling and root planing in the treatment of periodontal inflammation. *J. Periodontol.* **81** (8), 1161–1166 (2010).
28. Üstün, K. et al. <article-title update="added">Clinical and Biochemical Effects of 810 nm Diode Laser as an Adjunct to Periodontal Therapy: A Randomized Split-Mouth Clinical Trial. *Photomed. Laser Surg.* **32**(2), 61–6 (2014).
29. Abduljabbar, T., Vohra, F., Kellesarian, S. V. & Javed, F. Efficacy of scaling and root planning with and without adjunct Nd: YAG laser therapy on clinical periodontal parameters and gingival crevicular fluid interleukin 1-beta and tumor necrosis factor-alpha levels among patients with periodontal disease: A prospective randomized split-mouth clinical study. *J. Photochem. Photobiol. B.* **169**, 70–4 (2017).
30. Dukić, W., Bago, I., Aurer, A. & Roguljić, M. Clinical effectiveness of diode laser therapy as an adjunct to non-surgical periodontal treatment: A randomized clinical study. *J. Periodontol.* **84**(8), 1111–1117 (2013).
31. Saglam, M., Kantarci, A., Dundar, N. & Hakki, S. S. Clinical and biochemical effects of diode laser as an adjunct to nonsurgical treatment of chronic periodontitis: A randomized, controlled clinical trial. *Lasers Med. Sci.* **29**(1), 37–46 (2014).
32. Makhlouf, M., Dahaba, M. M., Tunér, J., Eissa, S. A. & Harhash, T. A. H. Effect of adjunctive low level laser therapy (LLLT) on nonsurgical treatment of chronic periodontitis. *Photomed. Laser Surg.* **30** (3), 160–166 (2012).
33. Ren, C., McGrath, C., Jin, L., Zhang, C. & Yang, Y. The effectiveness of low-level laser therapy as an adjunct to non-surgical periodontal treatment: A meta-analysis. *J. Periodontal Res.* **52**(1), 8–20 (2017).
34. Moritz, A. et al. Treatment of periodontal pockets with a diode laser. *Lasers Surg. Med.* **22**(5), 302–11 (1998).
35. Lui, J., Corbet, E. & Jin, L. Combined photodynamic and low-level laser therapies as an adjunct to nonsurgical treatment of chronic periodontitis. *J. Periodontal Res.* **46** (1), 89–96 (2011).
36. Pangam, T. S. *Assessment of Effect of Diode Laser as an Adjunct to Scaling and Root Planing on Clinical Parameters and Colony Forming Units of Aggregatibacter Actinomycetemcomitans* (Rajiv Gandhi University of Health Sciences (India), 2018).
37. Yilmaz, S. et al. Effect of galium arsenide diode laser on human periodontal disease: a microbiological and clinical study. *Lasers Surg. Med. Off. J. Am. Soc. Laser Med. Surg.* **30**(1), 60–66 (2002).

38. De Micheli, G. et al. Efficacy of high intensity diode laser as an adjunct to non-surgical periodontal treatment: A randomized controlled trial. *Lasers Med. Sci.* **26**(1), 43–8 (2011).
39. Matarese, G., Ramaglia, L., Cicciù, M., Cordasco, G. & Isola, G. The effects of diode laser therapy as an adjunct to scaling and root planing in the treatment of aggressive periodontitis: A 1-year randomized controlled clinical trial. *Photomed. Laser Surg.* **35**(12), 702–709 (2017).

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Author contributions
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Declarations

Competing interests
The authors declare no competing interests.

Ethics approval and consent to participate
The study protocol was approved by the University Research Ethics Committee (UREC) of the University of Science and Technology (UST), Sana'a, Yemen (Study ID: 1446/0016/UREC/UST). All methods were performed in accordance with the declaration of Helsinki.

Additional information
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Correspondence and requests for materials should be addressed to A.A.A.-S.

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